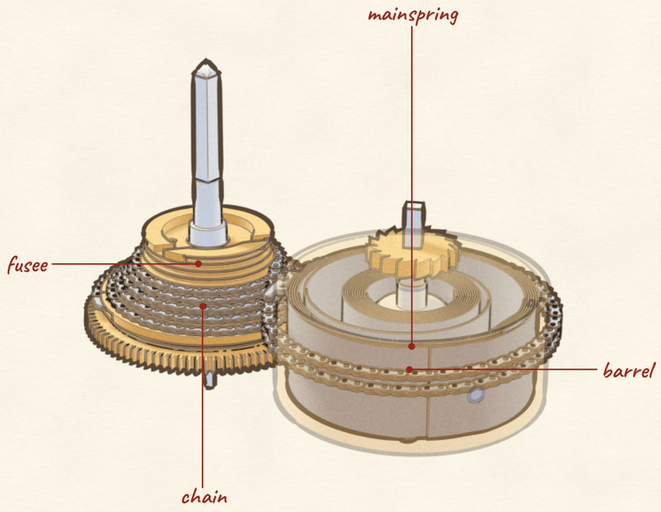
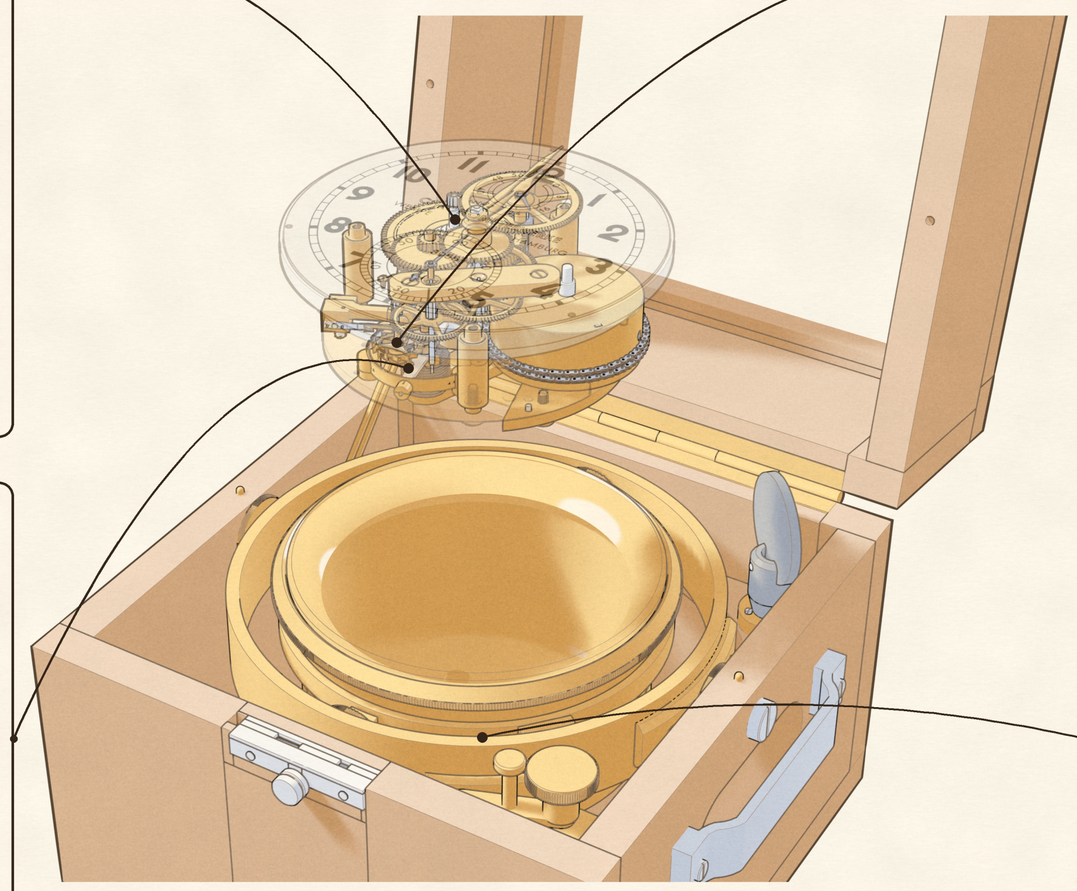
What makes a marine chronometer precise

The Wempe Einheits-Chronometer, drawn from its working model: every line is the model's, built from Wempe's works drawings

1 An even pull: the fusee

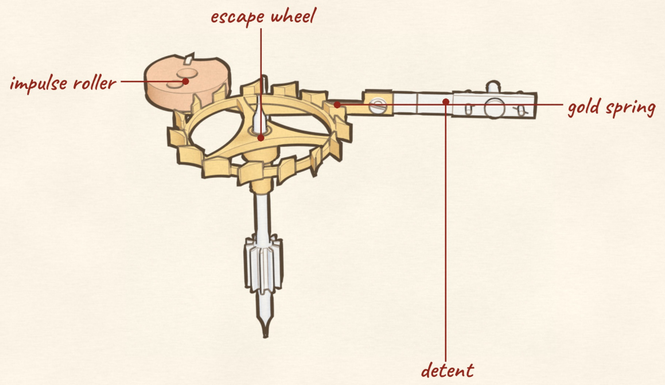
A mainspring pulls hardest fully wound. A fine steel chain carries its pull to the fusee, a grooved cone: wound up, the chain pulls on the cone's narrow end, and as the spring weakens it pulls at larger and larger radii, so the force reaching the train stays nearly even. 8¾ turns of the fusee run the Wempe its 56 hours.





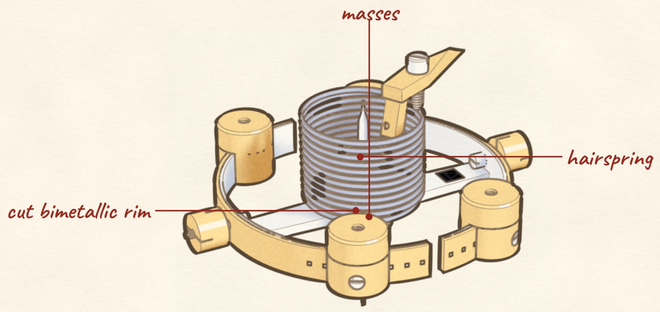
2 Touching the balance as little as it can

The detent, a slender blade on a thin spring, holds the escape wheel, 15 teeth and 14 mm across. Swinging one way, the balance's discharge stone lifts it through the gold spring; a tooth then drives the impulse roller, and the detent springs back to lock the next. Swinging back, the gold spring just bends aside: the balance swings almost free.



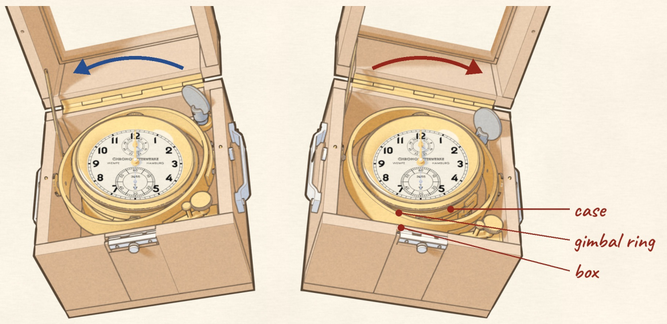
3 Heat and the balance

Warmed, a steel spring weakens and a balance grows, and both slow the clock. The Wempe's rim is nickel steel and brass, cut through at 90° from the arm: brass expands more, so each quarter's free end curls inward, drawing mass toward the axis. The brass masses' place on the rim sets how much.



4 Level on a rolling ship

The case hangs in a ring pivoted in the box, the ring on pivots at right angles to the case's. Its weight hangs low under the pivots, so it stays level while the box rolls and pitches round it: its own swing takes under a second, a ship's roll ten or more.



the box rolled one way: the case level

and the other: the case level

Sources: Wempe's works drawings (dg-chrono.de) and the working model at marinechronometermodel.com, where each part can be turned, taken apart and set running.